

STAT C1000 — Exam 1 Practice Answer Key

Version P1 · Model answers (English) + brief notes (日本語 OK)

Question 1 — Variable classification (7 pts each)

Write-up format: (1) variable · (2) Qual/Quant · (3) subtype · (4) level

A) Variable: number of streaming-app subscriptions

Quantitative · Discrete · Ratio

注: 数えられる個数。0は「契約なし」の真のゼロ → discrete + ratio。

B) Variable: time of day first class begins (military / 24-hour clock)

Quantitative · Continuous · Interval

注: 時刻。差は意味あるが「0:00」は「時間がない」ではない → interval (military time の罫) 。

C) Variable: laptop weight (pounds)

Quantitative · Continuous · Ratio

注: 測れる量で真のゼロあり (0 lb = 重さなし) → continuous + ratio。

D) Variable: customer satisfaction star rating (1–5)

Qualitative · Ordinal · Ordinal

注: 星は順序ラベル。数値に見えるがカテゴリ → qualitative + ordinal。

E) Variable: walking time from parking lot to classroom

Quantitative · Continuous · Ratio

注: 所要時間 (duration) 。0は本当のゼロ、比も意味あり → continuous + ratio。

Question 2 — SMC paid work hours by major (24 pts)

A) Parameter: μ = the true mean number of hours SMC students spend working at a paid job per week.

(Symbol 2 pts + description 2 pts)

注: 「true average / true mean」なら母平均 μ 。標本平均は $\bar{x}$ なので混同しない。

B) All Santa Monica College students (or: the entire SMC student body). (2 pts)

注: 対象母集団は「SMCの全学生」。標本の300人ではない。

C) Variable: number of hours spent working at a paid job per week. Quantitative, continuous, ratio level of measurement. (Description 4 pts + type/level 4 pts)

注: 時間は測れる連続量。0時間は本当のゼロ → ratio。 (major は層分けの軸であり、関心変数ではない)

D) Sample: 300 students. Sampling method: stratified random sampling (strata = major; random sample from each major). (3 pts)

注: majorで分けて各層からrandom = stratified。clusterなら「いくつかのmajorを選び、その中の全員」になる。

E) Statistic: $\bar{x}$ (sample mean hours worked per week). Calculate: $\bar{x} = (\text{sum of the 300 students' weekly paid-work hours}) / 300$. (Symbol 4 pts + calculation 3 pts)

注: 統計量は標本から計算。平均なら合計÷n。

Quick traps checklist

- military / 24-hour clock time → Interval (not ratio)
- star / Likert ratings → Qualitative + Ordinal
- counts of things → Discrete + Ratio
- weight / duration / money / hours worked → Continuous + Ratio

- break into groups, sample from each → Stratified (not cluster)